

Harsh Sharma

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Professional Summary

Computer Science M.S. student focused on building agentic AI systems, observability infrastructure, and ML-backed software systems, with hands-on experience across backend services, ML pipelines, and cloud-native infrastructure.

Education

Arizona State University 08/2025 - 05/2027

Master of Science in Computer Science — Tempe, AZ

GPA: 3.89/4.00 | Coursework: Machine Learning, Artificial Intelligence, Data Mining, Deep Learning, Knowledge Representation

Netaji Subhas University of Technology (NSUT) 11/2021 - 05/2025

Bachelor of Engineering in Electronics and Communication Engineering — New Delhi, India

GPA: 8.32/10.0 | Coursework: Data Structures, Probability, Statistics, Machine Learning, Digital Signal Processing

Technical Skills

Languages & Systems: Python, C++, Go, SQL, Bash, Linux, Git

Backend & Cloud Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, Helm, GitHub Actions, Service-Oriented Design

Observability & Reliability: OpenTelemetry, OpenObserve, OTLP, Distributed Tracing, Metrics, Structured Logging, Drift Monitoring

Agentic AI & ML: LangGraph, LLM Tool Calling, RAG, Vector Retrieval, PyTorch, TensorFlow, scikit-learn, Transformers, Anomaly Detection

Data & Evaluation: NumPy, Pandas, ChromaDB, PM4Py, Evaluation Pipelines, Experiment Tracking

Experience

Arizona State University 08/2026 - Present

Agentic AI Guide & Course Developer — Semantic Web Mining — Tempe, AZ

- Designed and built Agent Harness, a self-built multi-agent orchestration platform (LangGraph, FastAPI, Streamlit), from scratch as the course's hands-on teaching tool for agentic AI system design (full architecture detailed under Selected Projects).
- Authored course materials, architecture documentation, and cross-platform setup guides, and delivered lectures introducing students to agentic AI concepts, LangGraph orchestration, and observability.
- Graded student assignments and projects, providing technical feedback on agentic system design, reasoning-loop correctness, and debugging practices.

VISA Lab, Arizona State University 04/2026 - 07/2026

Graduate Researcher – Process Mining & Agentic AI Systems — Tempe, AZ

- Developed a process-mining digital twin over BPIC20 event logs, exposing event, resource, process, and historical-resolution context through structured tool interfaces for evidence-grounded LLM agent analysis.
- Designed a LangGraph-based workflow integrating investigator, recommendation, explanation, and evaluation stages with PM4Py process models and replay-based what-if analysis for corrective-action ranking.

Sysmat Research Solutions Pvt. Ltd. | Remote 06/2024 - 08/2024

Machine Learning Engineer Intern – ML Infrastructure & Signal Pipelines

- Built modular Python pipelines for preprocessing 64-channel EEG datasets and implemented dynamic neural-graph generation with spatio-temporal GCN-GRU workflows for multichannel classification, improving validation accuracy by 9%.
- Orchestrated preprocessing, model-input generation, evaluation, and structured logging through reusable workflows, reducing experimentation and handoff time by 40%.

Bow Creek Financial | Remote 06/2023 - 08/2023

Machine Learning Engineer Intern – Time-Series & Anomaly Detection

- Implemented Python pipelines for financial time-series ingestion and feature engineering, integrating ARIMA and autoencoder models with automated threshold tuning, improving rare-event recall by 22%.
- Built reproducible training, inference, and evaluation workflows with Docker, structured metric logging, and reusable experiment artifacts.

Selected Projects

[Agent Harness – Modular Multi-Agent Orchestration System](#)

- Built a modular multi-agent orchestration harness with LangGraph, FastAPI, and Streamlit, featuring a Base Orchestrator that routes requests to specialized agents through isolated private execution graphs.
- Designed a Text2SQL agent with bounded schema discovery, deterministic SQL safety validation, and iterative multi-loop reasoning, using structured Pydantic outputs and LangSmith tracing for full observability.
- Containerized the system with Docker Compose and authored cross-platform setup scripts for reproducible deployment.

[RepoScope – Repository-Aware Code Analysis and Review System](#)

- Built a repository-aware AI backend that indexes source code into a persistent vector database and retrieves relevant cross-file context for LLM-based code review, using semantic chunking, bounded context expansion, and FastAPI services separating indexing, retrieval, and generation.